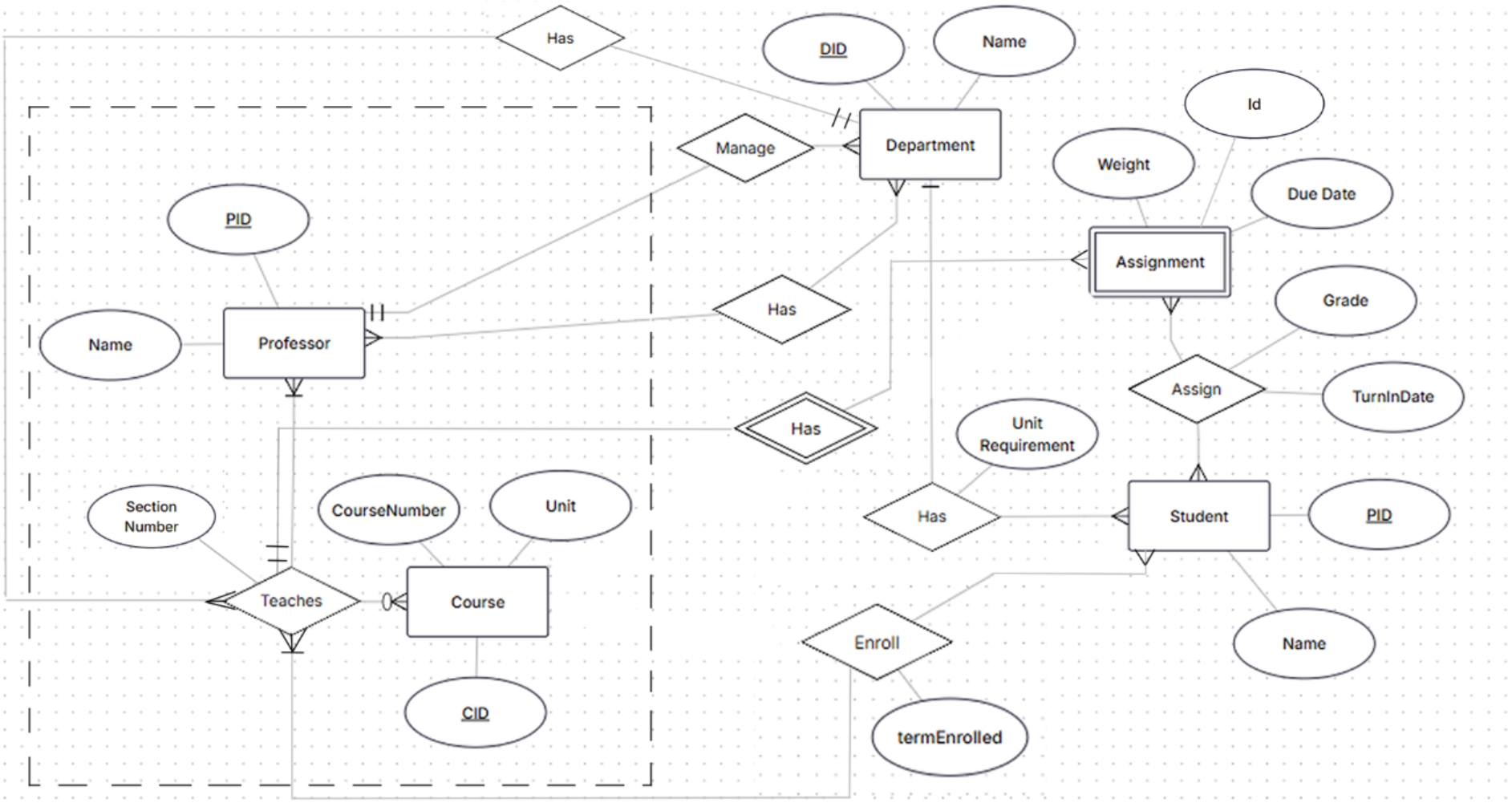
CS166 Grade Database - ER Diagram

Original team diagram. The following page records the clarifications used by the SQL.



Original image preserved from repository commit 974905d. See docs/relational-schema.md for keys and constraints.

Implementation clarifications

Student identifier

The Student identifier labeled PID in the image is SID (sid in SQL).

Section identity

Teach uses K = (pid, cid, sectionnumber, term). The term is part of the offering, so the same section number can be reused across terms. dept is an attribute determined by K, not part of its primary key.

Assignment identity

Assignment is identified by (K, id). Assign is identified by (K, id, sid), so each result identifies both an assignment and an enrolled student. Enroll is identified by (K, sid).

Term attribute

termEnrolled is represented by the term in K rather than repeated as an enrollment-only non-key attribute. Dependent keys carry the same offering term.

Departments and professors

WorksIn implements professor-department membership. Each department manager and each section professor must have the corresponding membership. A department has exactly one manager; a professor may manage multiple departments.

Draft participation

Sections may initially have no enrollments or assignments; assignments may initially have no results. Each stored dependent row must reference its parent. These optional parent-side cases allow records to be created before teaching/grading begins.

Grades and weights

Grades and weights are decimal percentages. Assignment IDs serve as labels. Grades and submission/due dates may be unknown. A zero grade is distinct from an unrecorded grade. Weight totals of 100 are required before finalization; draft schemes may be incomplete.

Scope

The original conceptual entities and associations are retained. This page records implementation decisions and does not claim instructor approval. Full requirements, keys, and functional dependencies are in docs/requirements.md and docs/relational-schema.md.